

Interpretable Heart Disease Classification: A Comparative Study of Six Machine Learning Classifiers Using SMOTETomek and SHAP

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Abstract—Heart disease remains a major public health concern, motivating the development of machine learning methods that can identify patterns associated with reported heart disease status. This study presents a comparative evaluation of six machine learning classifiers using demographic, behavioral, and health-related features from the 2023 BRFSS LLCP2023.XPT dataset. Because the target classes are imbalanced, SMOTE-Tomek is applied to the training data to address class imbalance while preserving informative samples. Six classification models are evaluated using accuracy, precision, recall, F1-score, and ROC-AUC, with particular attention to recall because false-negative predictions are important in health-related classification tasks. Five-fold stratified cross-validation, with resampling performed within each training fold, is used to assess the stability of model performance. In addition to predictive performance, SHAP (SHapley Additive exPlanations) is used to interpret model predictions and visualize the contribution of individual features, providing a more accessible representation of model behavior for non-technical healthcare stakeholders. The results demonstrate differences between models across evaluation metrics and show that higher overall accuracy does not necessarily correspond to higher recall. SHAP analysis identifies factors such as age group, self-reported general health, and high blood pressure status among the influential features in the model's predictions. These findings highlight the importance of evaluating both predictive performance and interpretability when applying machine learning to health-related classification problems.

Index Terms—Heart disease classification, machine learning, XGBoost, random forest, SMOTETomek, SHAP, class imbalance, healthcare AI

I. INTRODUCTION

Advances in artificial intelligence (AI) and machine learning (ML) have enabled healthcare systems to analyze medical data and identify patterns associated with cardiovascular disease. Prior studies have demonstrated the potential of machine learning for heart disease prediction and early-stage classification [1], [2]. By analyzing large volumes of patient data—including age, blood pressure, cholesterol, prior diagnoses, and behavioral factors—ML models can identify patterns associated with reported heart disease status that support, rather than replace, clinical judgment.

This study builds a classification pipeline using the 2023 BRFSS LLCP2023.XPT dataset (hereafter referred to as LLCP2023) and systematically compares six classification approaches across multiple evaluation metrics, while incorporating SHAP-based explainability to make model predictions

more interpretable to non-technical healthcare stakeholders. Rather than reporting a single accuracy figure, this work treats accuracy, F1-score, area under the receiver operating characteristic curve (ROC-AUC), and recall on the positive class as complementary lenses, since on an imbalanced medical dataset the metric that performs best is not necessarily the most informative metric for evaluating minority-class identification.

The main objectives of this research are:

- Construction of a reproducible heart-disease classification pipeline from LLCP2023 survey file.
- Comparative evaluation of six ML algorithms under identical preprocessing and evaluation conditions.
- Mitigation of severe class imbalance using SMOTE-Tomek, applied without leakage into the test set.
- Explanation of model predictions using SHapley Additive exPlanations (SHAP), to identify which demographic, health-related, and behavioral features contribute to model predictions.

II. LITERATURE REVIEW

Machine learning has been increasingly applied to heart disease prediction and medical data analysis. Previous studies have explored different classification algorithms, feature-selection strategies, optimization techniques, and application-oriented prediction systems [1]– [7]. Explainable artificial intelligence (XAI) has emerged as an important approach for improving the transparency and interpretability of machine learning predictions in healthcare. SHAP provides a unified framework for explaining individual model predictions by attributing contributions to input features [8]. In healthcare settings, such explanations may help make model outputs more understandable to clinical users; however, evidence regarding the effect of XAI on clinician trust is mixed, suggesting that the clarity and quality of explanations are important considerations when designing explainable healthcare AI systems [9]. A limitation in evaluating imbalanced classification problems is that accuracy alone can obscure minority-class performance, since a classifier can achieve high accuracy by favoring the majority class. This study addresses this issue by reporting a majority-class baseline alongside the trained classifiers and evaluating accuracy together with precision, recall, F1-score,

TABLE I
SOFTWARE LIBRARIES AND TOOLS USED

Library / Tool	Purpose in This Project
Python 3.x	Primary language for all data processing, model training, and evaluation
pandas / NumPy	Loading and manipulating the LLC2023 XPT transport file; numerical array operations
scikit-learn	Stratified train/test split, StandardScaler, SimpleImputer, all evaluation metrics
XGBoost	Primary classification model: 200-estimator gradient boosted tree ensemble
imbalanced-learn	SMOTETomek resampling, which combines SMOTE oversampling with Tomek-link under-sampling
SHAP	Model explainability: TreeExplainer computes Shapley values for XGBoost
Matplotlib / Seaborn	Confusion matrix heatmap and chart visualization
Google Colab	Compute environment for running XGBoost training with <code>n_jobs=-1</code>

and ROC-AUC, with particular attention to minority-class recall. Accordingly, this study combines SMOTETomek-based resampling, comparative evaluation across six classifiers, and SHAP-based analysis of model predictions.

III. METHODOLOGY

Table I lists the software used in this work.

A. Dataset

The 2023 BRFSS combined landline and cellular telephone data file (LLCP2023), published by the Centers for Disease Control and Prevention (CDC), was used to train and evaluate all models. The raw file contains 433,323 records and 350 variables. After dropping rows with a missing target label, 428,738 records remained (392,427 negative, 36,311 positive). No duplicate rows were found; seven duplicate columns (WKENDOUT, SUNPRCT, CTELENUM1, CADULT1, NUMBURN3, WKDAYOUT, CELLFON5) were identified and removed to improve data quality prior to training. The target variable `_MICH` was converted to a binary label:

- 1 = coronary heart disease (CHD) or myocardial infarction (MI) ever reported
- 0 = neither CHD nor MI reported

The cleaned dataset was downsampled to a fixed, reproducible sample of 80,000 records (`random_state = 42`), yielding a class split of 73,280 negative (91.6%) and 6,720 positive (8.4%) cases, closely preserving the class proportions of the full file. The label reflects respondents' self-reported history rather than clinical records, and the BRFSS survey weights were not applied, so the results describe the sampled respondents rather than population-level prevalence.

B. Feature Selection

Twenty-six predictors were selected using clinical and behavioral domain knowledge rather than automated correlation filtering, spanning self-rated health, cardiovascular and

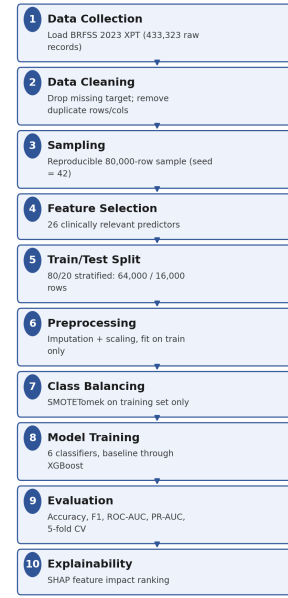


Fig. 1. Process diagram of the heart disease prediction pipeline.

chronic-condition history, lifestyle factors, demographics, socioeconomic status, and functional limitations. Selected variables include general health, physical and mental health days, high blood pressure, high cholesterol, prior stroke, diabetes, chronic obstructive pulmonary disease, depressive disorder, kidney disease, arthritis, asthma, smoking status, alcohol consumption, any exercise and strength-training frequency, body mass index (BMI), age group, sex, education, income, employment, and difficulty dressing, walking, or making decisions. The most incomplete predictors were average drinks per day (`AVEDRNK3`, 40,653 missing), high cholesterol status (`TOLDHI3`, 9,507 missing), and BMI (`_BMI5`, 7,284 missing).

C. Preprocessing

The overall workflow is summarized in Fig. 1. Following an 80/20 stratified train/test split (64,000 training rows, 16,000 test rows, preserving the 91.6%/8.4% class ratio in both sets), the preprocessing pipeline applied:

- Missing-value imputation using the most-frequent-value strategy, fit only on the training set.
- Feature scaling via `StandardScaler`, fit only on the training set and applied unchanged to the test set.
- Class balancing via `SMOTETomek`, applied only to the training set after scaling.

Fitting the imputer and scaler exclusively on training data, then applying the learned transformation to the test set, avoids information leakage from the test set into preprocessing decisions.

D. Class Imbalance Handling

`SMOTETomek` combines the Synthetic Minority Over-sampling Technique (SMOTE), which synthesizes new minority-class samples by interpolating between existing ones,

with Tomek-link undersampling, which removes ambiguous majority-class samples near the decision boundary. Applied to the 64,000-row training set (58,624 negative, 5,376 positive), this produced a balanced training set of 117,128 rows (58,564 per class). This was applied exclusively to the training set; the test set retained the class distribution of the sampled data (14,656 negative, 1,344 positive) so that reported metrics are not inflated by an artificially balanced evaluation.

E. Machine Learning Algorithms

Six classifiers were trained and evaluated under identical preprocessing, all with a random seed of 42:

- Baseline (majority-class dummy classifier), trained on the original, unbalanced training labels to establish a sanity-check floor.
- Logistic regression (maximum of 1,000 iterations).
- Decision tree (maximum depth 6).
- Random forest (200 trees, maximum depth 8).
- Support vector machine (SVM, radial basis function kernel), trained on a reproducible 20,000-row subsample of the resampled training set due to the computational cost of kernel SVM on large datasets.
- XGBoost (extreme gradient boosting; 200 estimators, maximum depth 4, learning rate 0.1).

All models except the baseline were trained on the SMOTETomek-resampled training data and evaluated on the untouched, imbalanced test set (16,000 rows: 14,656 negative, 1,344 positive).

F. Validation Strategy

To assess whether the single-split results were sensitive to one particular train/test partition, 5-fold stratified cross-validation (shuffled, seed 42) was performed on the 64,000-row training partition for logistic regression, decision tree, random forest, and XGBoost, using accuracy, F1-score, and ROC-AUC as evaluation metrics. SMOTETomek was applied within an `imbalanced-learn` pipeline for each fold, ensuring that resampling was performed only on the corresponding training fold. The imputer and scaler were fit once on the full training partition before cross-validation and then used throughout the folds; the held-out test set was not used during cross-validation.

IV. RESULTS AND ANALYSIS

A. Model Performance Comparison

Table II reports accuracy and, for the positive (heart-disease) class, precision, recall, and F1-score, together with ROC-AUC, for all six models on the held-out 16,000-row test set.

Among the trained models, XGBoost achieves the highest accuracy and ROC-AUC, although its accuracy is still marginally below the 91.60% of the majority-class baseline. Its F1-score on the positive class (0.195) is the lowest among the five non-baseline models. This is explained by its low recall: only 12.2% of true positive cases are correctly identified, versus a precision of 48.0% on the cases it does flag. In effect, XGBoost’s high accuracy is driven largely by its strength

TABLE II
MODEL PERFORMANCE COMPARISON ON THE HELD-OUT TEST SET
($n = 16,000$)

Model	Acc. (%)	Prec.	Rec.	F1	AUC
Baseline	91.60	0.0000	0.0000	0.0000	—
Log. regression	74.05	0.2137	0.7798	0.3355	0.8348
Decision tree	81.55	0.2467	0.5826	0.3466	0.8031
Random forest	83.67	0.2764	0.5833	0.3751	0.8389
SVM	77.76	0.2293	0.6979	0.3452	0.8204
XGBoost	91.51	0.4795	0.1220	0.1945	0.8445

TABLE III
5-FOLD STRATIFIED CROSS-VALIDATION RESULTS

Model	Acc. mean (%)	Acc. std (pp)	F1	AUC
Log. regression	73.32	0.47	0.330	0.8298
Decision tree	81.68	1.30	0.340	0.8016
Random forest	83.11	0.13	0.363	0.8309
XGBoost	91.59	0.15	0.211	0.8371

on the majority (no-disease) class, since it closely tracks the 91.6% baseline accuracy of predicting “no disease” outright. Random forest, by contrast, trades some accuracy (83.67%) for substantially higher recall (58.3%), yielding the best F1-score (0.375) of any model tested. Logistic regression attains the highest recall (78.0%) but at the lowest precision among the trained models (21.4%). These results demonstrate that the model with the highest overall accuracy is not necessarily the model with the highest minority-class recall or F1-score, highlighting the importance of selecting evaluation metrics according to the intended classification objective.

B. Cross-Validation Results

Table III reports 5-fold stratified cross-validation results (mean and standard deviation of accuracy; mean F1 and ROC-AUC) with SMOTETomek applied inside the pipeline per fold.

Cross-validation results closely track the single-split results in Table II, with low standard deviations across folds (0.13–1.30 percentage points on accuracy), confirming that the single-split comparison is stable and not the result of a favorable random partition.

C. SHAP Explainability Analysis

SHAP TreeExplainer values were computed for the XGBoost model on the first 1,000 rows of the scaled test set (Fig. 2). The features with the greatest mean impact on model output, in descending order, were:

- Age group (`_AGE_G`), the single strongest predictor by a clear margin
- Self-reported general health (`GENHLTH`)
- High blood pressure status (`BPHIGH6`)
- Income bracket (`INCOME3`)
- High cholesterol status (`TOLDHI3`)
- Sex (`SEXVAR`)

These findings provide an interpretable view of the features that contributed most strongly to the XGBoost model’s predictions.

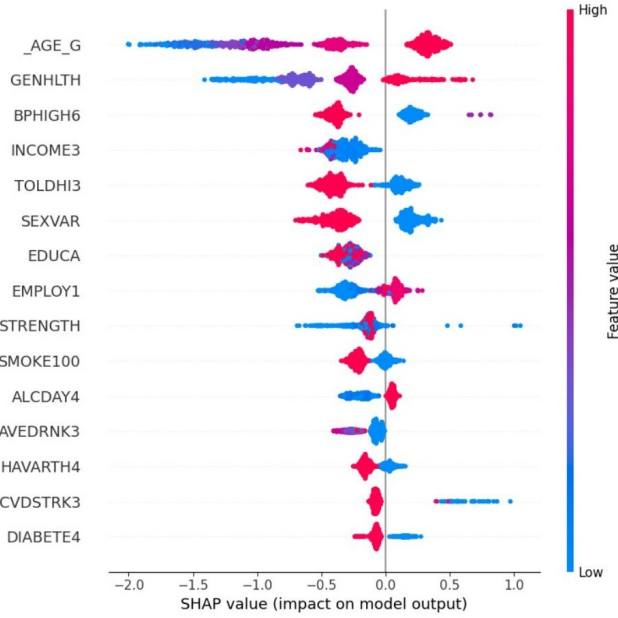


Fig. 2. SHAP summary plot: feature impact on XGBoost output.

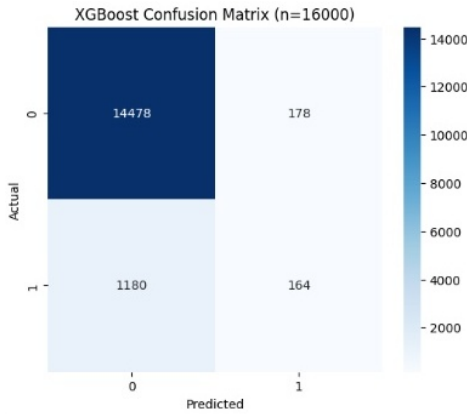


Fig. 3. XGBoost confusion matrix ($n = 16,000$).

D. Confusion Matrix Analysis (XGBoost)

Fig. 3 shows XGBoost’s confusion matrix on the 16,000-row test set: 14,478 true negatives, 178 false positives, 1,180 false negatives, and 164 true positives. The high false-negative count is the direct, numerical expression of the model’s low recall (12.2%). This means roughly seven out of every eight true heart-disease cases in this sample would be missed if XGBoost’s raw class predictions were used without adjustment (e.g., threshold tuning or recall-oriented model selection).

Fig. 4 summarizes XGBoost’s key metrics on the same test set, including weighted and macro F1-scores alongside accuracy, ROC-AUC (0.8445), and precision–recall AUC (PR-AUC, 0.3330), together giving a fuller picture than accuracy alone. The PR-AUC is roughly four times the 0.084 positive-class prevalence, which is the PR-AUC expected from a classifier with no skill. The sharp drop from weighted F1 (89.13%)

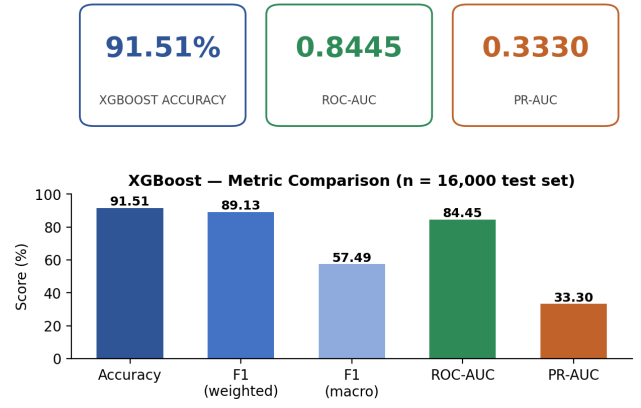


Fig. 4. XGBoost model accuracy summary ($n = 16,000$ test set).

to macro F1 (57.49%) is itself informative: it reflects the same class-imbalance effect discussed throughout this paper, since weighted F1 is dominated by the majority class, while macro F1 weights both classes equally and therefore exposes the model’s weaker performance on the minority (heart-disease) class.

V. DISCUSSION

This study underscores that accuracy alone is an unreliable, and potentially misleading, metric for imbalanced medical classification tasks. Because only 8.4% of the sampled respondents report a history of CHD or MI, a model can exceed 91% accuracy while recognizing very few true positive cases, as XGBoost’s 12.2% recall demonstrates. Reporting a majority-class baseline (91.60% accuracy, 0.000 F1) alongside trained models makes this risk explicit and prevents an inflated sense of model quality.

Random forest’s higher F1-score indicates a more balanced precision–recall profile than XGBoost at the default decision threshold in this experiment. This illustrates how model selection can depend on the evaluation metric and classification objective rather than accuracy alone. This trade-off could be further managed in future work through threshold tuning, cost-sensitive learning, or recall-oriented hyperparameter search on XGBoost itself, rather than treating the six models as fixed, mutually exclusive choices.

SHAP explainability analysis adds interpretive value beyond raw metrics by showing how individual features contributed to the XGBoost model’s predictions. Compared with prior work in this space, this study’s contributions include:

- An explicit majority-class baseline to contextualize all reported accuracy figures.
- Preprocessing and resampling that never use the held-out test set: imputation, scaling, and SMOTETomek are fit only on training data.
- A six-model comparison spanning linear, tree-based, ensemble, kernel, and boosting approaches under identical conditions.
- Cross-validated confirmation of single-split findings.

- SHAP-based explainability tied to the highest-accuracy trained model, plus a candid discussion of its recall limitation.

VI. CONCLUSION

This study proposed and evaluated a heart-disease classification framework using the 2023 BRFSS LLCP2023 data file across six classification models. Among the trained models, XGBoost recorded the highest accuracy (91.51%) and ROC-AUC (0.8445), while random forest recorded the highest minority-class F1-score (0.375). Five-fold cross-validation confirmed these results were stable across data partitions. SHAP analysis identified age group, general health, and blood pressure status as the features with the greatest influence on the XGBoost model's predictions.

The results demonstrate that model evaluation for imbalanced health-related classification should consider the relative importance of false-negative errors and should not rely on accuracy alone. Future work may include threshold tuning and cost-sensitive learning to improve recall without sacrificing the interpretability and reliability demonstrated here, as well as extending this framework to other chronic disease prediction tasks and exploring deployment through a lightweight web interface for research and public-health screening applications.

Future directions include:

- Recall-oriented threshold tuning and cost-sensitive learning.
- Extension of this framework to other chronic disease prediction tasks.
- Development of a web-based interface for accessible demonstration and screening of model predictions.

ACKNOWLEDGMENT

The author sincerely thanks the Department of Computer Science and Engineering, Adamas University, for their guidance and support throughout this project.

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